

DevStore Enterprise Platform

Full Project Architecture, Module Breakdown & Technical Documentation

1. Executive Summary & Project Purpose

DevStore is an enterprise-grade, full-stack e-commerce application designed with a sleek dark-mode glassmorphic user interface. It combines a seamless consumer shopping storefront with a highly secure Admin Console for product catalog management, order fulfillment tracking, user administration, and automated logistics updates.

Key System Features & Capabilities:

- > **Dedicated Admin Security Gateway (/admin):** Restricted entry point at /admin requiring explicit administrator credentials, bypassing public redirects.
- > **Global Address Autocomplete:** Dynamic location predictions on Checkout address fields using OpenStreetMap / Nominatim API.
- > **Enterprise Invoice PDF Generator:** Vector PDF billing invoices with dynamic, non-overlapping financial breakdown coordinates.
- > **Automated 24-Hour Shipping Transition:** Backend lifecycle checker that automatically transitions processing orders to shipped status after 24 hours.
- > **Real-Time Dynamic Data Syncing:** Smooth background polling without screen flashing or skeleton loading spinner resets.

2. System Architecture & Technology Stack

The repository is structured as a Monorepo containing separate frontend (client/) and backend (server/) workspaces.

Layer	Technologies Used	Purpose & Easy Explanation
Frontend UI	React 18, Vite, Tailwind CSS, Lucide Icons	Renders fast interactive pages with modern glassmorphism styling.
State Management	React Context API (AppContext)	Handles global user sessions, shopping cart, and notification toasts.
Backend API	Node.js, Express.js, JWT, Middleware	Serves RESTful endpoints, handles auth verification, and handles file uploads.
Database Tier	SQLite / Local JSON Collections (jsonDb)	Zero-setup dynamic database system for immediate local data persistence.

3. Core Application Modules & Easy Explanations

1. Dedicated Admin Gateway (/admin)

When visiting /admin, the app presents a high-security dark glass card asking for Admin Email and Password. It includes a 1-click Auto-fill button for admin@devstore.com / Admin123!. Once authenticated, it unlocks full administrative controls on the same page.

2. User Management (Read-Only Table)

Renders all registered users in a clean, read-only table displaying Name, Email, and Role badge (ADMIN / USER). All action buttons and role toggles were removed to enforce strict read-only security. Non-admin users are strictly blocked by server-side middleware.

3. Address Autocomplete on Checkout

When filling shipping details at Checkout, typing in Street Address, City, State, or Country triggers a pop-down list of real location suggestions powered by OpenStreetMap/Nominatim API. Clicking any suggestion auto-fills the entire address.

4. Enterprise PDF Invoice Generator

Clicking "Invoice PDF" streams a professional vector PDF document. The financial summary (Subtotal, Shipping, Tax included, Discounts, Grand Total) features dynamic coordinate positioning, ensuring zero overlapping text labels.

5. Automated 24-Hour Shipping Lifecycle

Whenever orders are fetched, the server compares Date.now() against the order's createdAt timestamp. If 24 hours or more have passed since order placement, the order status automatically transitions from PROCESSING to SHIPPED in the database.

4. Core REST API Endpoint Reference

Method	Endpoint Route	Access Level	Functionality & Easy Explanation
POST	/api/auth/login	Public	Authenticates user credentials and issues JWT token.
POST	/api/auth/register	Public	Creates new user account with encrypted password.
GET	/api/products	Public	Fetches catalog products with search & category filters.
GET	/api/admin/users	Admin Only	Fetches registered user accounts (strictly protected).
GET	/api/admin/orders	Admin Only	Fetches orders with automated 24-hour shipping logic.
GET	/api/orders/invoice/:id	Authenticated	Generates and streams the Enterprise PDF Invoice.

Project Health & Verification Summary

The DevStore application has been fully developed, debugged, and verified. Both frontend and backend components operate without errors, providing a robust, high-performance E-Commerce platform.

